connection with aggressive contests between individual animals. I therefore postpone further discussion of this point until the end of that chapter, and return to the central message of this one. This is that the basic unit of natural selection is best regarded not as the species, nor as the population, nor even as the individual, but as some small unit of genetic material which it is convenient to label the gene. The cornerstone of the argument, as given earlier, was the assumption that genes are potentially immortal, while bodies and all other higher units are temporary. This assumption rests upon two facts: the fact of sexual reproduction and crossing-over, and the fact of individual mortality. These facts are undeniably true. But this does not stop us asking why they are true. Why do we and most other survival machines practise sexual reproduction? Why do our chromosomes cross over? And why do we not live for ever?

The question of why we die of old age is a complex one, and the details are beyond the scope of this book. In addition to particular reasons, some more general ones have been proposed. For example, one theory is that senility represents an accumulation of deleterious copying errors and other kinds of gene damage which occur during the individual's lifetime. Another theory, due to Sir Peter Medawar, is a good example of evolutionary thinking in terms of gene selection.* Medawar first dismisses traditional arguments such as: 'Old individuals die as an act of altruism to the rest of the species, because if they stayed around when they were too decrepit to reproduce, they would clutter up the world to no good purpose.' As Medawar points out, this is a circular argument, assuming what it sets out to prove, namely that old animals are too decrepit to reproduce. It is also a naïve group selection or species selection kind of explanation, although that part of it could be rephrased more respectably. Medawar's own theory has a beautiful logic. We can build up to it as follows.

We have already asked what are the most general attributes of a 'good' gene, and we decided that 'selfishness' was one of them. But another general quality that successful genes will have is a tendency to postpone the death of their survival machines at least until after reproduction. No doubt some of your cousins and great-uncles died in childhood, but not a single one of your ancestors did. Ancestors just don't die young!